

Assignment - 2

CRYPTOLOGY

1) encrypt "CHASE YOUR DREAMS" using

a) Caesar cipher

b) Shift cipher with key 7

c) ROT 13

⇒ a) Caesar cipher $k=3$

$$C = E(P+3) \text{ mod } 26 = F$$

⇒ CHASE

$$C :- E(2+3) \text{ mod } 26$$

$$5 \text{ mod } 26 \rightarrow 5 = F$$

$$H :- E(7+3) \text{ mod } 26$$

$$10 = K$$

$$A :- E(0+3) \text{ mod } 26$$

$$3 = D$$

$$S :- E(18+3) \text{ mod } 26$$

$$21 \text{ mod } 26 \rightarrow 21 = V$$

$$E :- E(4+3) \text{ mod } 26$$

$$7 = H$$

⇒ YOUR

$$Y = E(24+3) \text{ mod } 26$$

$$1 = B$$

$$O = E(14+3) \text{ mod } 26$$

$$17 \rightarrow R$$

$$U = E(20+3) \text{ mod } 26 \rightarrow 23 \rightarrow U$$

DREAMS:-

$$D: E(3+3) \bmod 26$$

$$6 \bmod 26 = 6 \rightarrow E$$

$$R = E(17+3) \bmod 26$$

$$20 = U$$

$$E = E(4+3) \bmod 26$$

$$7 = H$$

$$A = E(0+3) \bmod 26$$

$$3 = D$$

$$S = E(18+3) \bmod 26$$

$$21 = V$$

FLKDVH BRXU GUHOPV

b) Shift cipher $k=7$

$$C = E(P+k) \bmod 26$$

CHASE

$$C: E(2+7) \bmod 26$$

$$9 \bmod 26 = 9 \rightarrow J$$

$$H: C = E(7+7) \bmod 26$$

$$14 \rightarrow O$$

$$A: C = E(0+7) \bmod 26$$

$$7 = H$$

classmate
Date _____
Page _____

$$S: C \equiv E(18+7) \pmod{26}$$

$$25 \pmod{26} = 25 \Rightarrow Z$$

$$E: C \equiv E(4+7) \pmod{26}$$

$$11 \Rightarrow L$$

→ YOUR:

$$Y: C \equiv E(24+7) \pmod{26}$$

$$5 \Rightarrow F$$

$$O: C \equiv E(14+7) \pmod{26}$$

$$21 \Rightarrow V$$

$$U: C \equiv F(20+7) \pmod{26}$$

$$1 \Rightarrow B$$

$$R: C \equiv E(17+7) \pmod{26}$$

$$24 \Rightarrow Y$$

→ DREAMS:

$$D: C \equiv E(3+7) \pmod{26}$$

$$10 \Rightarrow K$$

$$R: C \equiv E(17+7) \pmod{26}$$

$$24 \Rightarrow Y$$

$$E \Rightarrow E(4+7) \pmod{26}$$

$$11 \Rightarrow L$$

$$A \Rightarrow E(0+7) \pmod{26}$$

$$7 \Rightarrow H$$

$$M: C = E(12+7) \bmod 26$$

$$19 \bmod 26 = 19 \rightarrow T$$

$$S: C = E(18+7) \bmod 26$$

$$= 25 \rightarrow Z$$

$\Rightarrow$ J O H Z L F V B Y K Y L H T Z

c) ROT 13 $K \rightarrow Z$

$$C = E(P+13) \bmod 26$$

$\Rightarrow$ CHASE :-

$$C: E(2+13) \bmod 26$$

$$15 \bmod 26 = 15 \rightarrow P$$

$$H: E(7+13) \bmod 26$$

$$20 \rightarrow U$$

$$A: E(0+13) \bmod 26$$

$$13 \rightarrow N$$

$$S: E(18+13) \bmod 26$$

$$5 \rightarrow F$$

$$E: E(4+13) \bmod 26$$

$$17 \rightarrow R$$

$\Rightarrow$ YOUR

$$Y: E(24+13) \bmod 26$$

$$11 \rightarrow L$$

$$O: E(14+13) \bmod 26 = 1 \rightarrow B$$

$$U: E(20+13) \bmod 26$$

$$7 \Rightarrow 4$$

$$R: C = E(17+13) \bmod 26$$

$$30 \bmod 26 = 4 \Rightarrow E$$

→ DREAMS:-

$$D: C = E(3+13) \bmod 26$$

$$16 \Rightarrow Q$$

$$R: E(17+13) \bmod 26$$

$$4 \Rightarrow E$$

$$E: E(4+13) \bmod 26$$

$$17 \Rightarrow R$$

$$A: E(0+13) \bmod 26$$

$$13 \Rightarrow N$$

$$M: E(12+13) \bmod 26$$

$$25 \Rightarrow Z$$

$$S: E(18+13) \bmod 26$$

$$5 \Rightarrow F$$

⇒ PUNFR LBHE QERNZF

2) Decrypt

3) "WRWDDOB VHFUHW PHVVDJH" using Caesar cipher

$$K=3$$

$$P = 10(1-3) \bmod 26 \Rightarrow F$$

$$W R W D O O B$$

$$W = (22-3) \bmod 26 \\ 19 \Rightarrow T$$

$$R = (17-3) \bmod 26 \\ 14 \Rightarrow O$$

$$W = (22-3) \bmod 26 \\ 19 \Rightarrow T$$

$$D = (3-3) \bmod 26 \\ 0 \Rightarrow A$$

$$O = (14-3) \bmod 26 \\ 11 \Rightarrow L$$

$$O \Rightarrow 11 \Rightarrow L$$

$$B = (1-3) \bmod 26 \\ 24 \Rightarrow Y$$

$$\Rightarrow V H F U H W$$

$$V = (21-3) \bmod 26 \\ 18 \bmod 26 = S$$

$$H = (7-3) \bmod 26 \Rightarrow E$$

$$F = (5-3) \bmod 26 \\ 2 \bmod 26 \Rightarrow C$$

$$U = (20 - 3) \bmod 26$$

$$17 \bmod 26 \Rightarrow R$$

$$H = (7 - 3) \bmod 26$$

$$4 \bmod 26 \Rightarrow E$$

$$W = (22 - 3) \bmod 26$$

$$19 \bmod 26 \Rightarrow T$$

$$\Rightarrow PHVDJH$$

$$P = (15 - 3) \bmod 26 \Rightarrow M$$

$$H = (7 - 3) \bmod 26$$

$$= E$$

$$V = (21 - 3) \bmod 26 = S$$

$$V \Rightarrow S$$

$$D = (3 - 3) \bmod 26 \Rightarrow A$$

$$J = (9 - 3) \bmod 26 \Rightarrow G$$

$$H = (7 - 3) \bmod 26$$

$$\Rightarrow E$$

$\Rightarrow$ TOTALLY SECRET MESSAGE

b) 'STJB BTWP YNRJ' using shift cipher 5

$$P = D(C - 5) \bmod 26$$

→ STB

$$S = (18 - 5) \bmod 26 \\ \Rightarrow N$$

$$J = (9 - 5) \bmod 26 \\ = E$$

$$B = (\overset{1}{8} - 5) \bmod 26 \\ = W$$

→ DTWP

$$D = (3-5) \bmod 26 \\ = 4$$

$$T = (19-5) \bmod 26 = 0$$

$$W = (22-5) \bmod 26 = R$$

$$P = (15-3) \bmod 26 \\ = 12$$

$$W = (22 - 5) \bmod 26$$

$\Rightarrow R$

$\Rightarrow YNRJ :-$

$$Y = (24 - 5) \bmod 26 = T$$

$$W = (13 - 5) \bmod 26 = I$$

$$R = (17 - 5) \bmod 26 = M$$

$$J = (9 - 5) \bmod 26 = E$$

~~$\Rightarrow$ NEW DWOOR TIME~~

$\Rightarrow$ NEW YORK TIME

c) FRAG ZBER PBSSRR using ROT 13
 $P = (C - 13) \bmod 26$

→ FRAG:

$$F = (5 - 13) \bmod 26 \\ = -8 \bmod 26 = 18 \rightarrow S$$

$$R = (17 - 13) \bmod 26 \\ \Rightarrow E$$

$$A = (0 - 13) \bmod 26 \\ \Rightarrow N$$

$$Q = (16 - 13) \bmod 26 \\ \Rightarrow O$$

→ ZBER:

$$Z = (25 - 13) \bmod 26 \Rightarrow M$$

$$B = (1 - 13) \bmod 26 \\ \Rightarrow O$$

$$E = (4 - 13) \bmod 26 \\ \Rightarrow R$$

$$R = (17 - 13) \bmod 26 = E$$

→ PBSSRR:

$$P = (15 - 13) \bmod 26 = C$$

$$B = (1 - 13) \bmod 26 = D$$

$$S = (18 - 13) \bmod 26 \Rightarrow F$$

$$S \Rightarrow F$$

$$R \Rightarrow (17 - 13) \bmod 26 \Rightarrow E$$

$$R \Rightarrow E$$

⇒ SEND MORE COFFEE